Customer Purchasing Behavior Report

1. Introduction

This report presents an in-depth examination of customer purchasing patterns derived from transactional data. The analysis is based on 3,900 recorded purchases across multiple product categories. The objective of this study is to identify meaningful trends in customer spending, segment behavior, product performance, and subscription impact in order to support data-driven business strategies.

2. Dataset Overview

Total Records: 3,900

Total Columns: 18

Main Data Categories

- Customer Information: Age, Gender, Location, Subscription Status
- Purchase Information: Product Name, Category, Purchase Amount, Season, Size, Color
- Behavioral Indicators: Discount Applied, Promo Code Usage, Previous Purchases, Frequency, Review Rating, Shipping Method

Data Quality: 37 missing values were identified in the Review Rating field and were handled during preprocessing.

3. Data Preparation and Cleaning (Python)

The dataset was imported using pandas. Initial structural checks were performed using df.info() and df.describe(). df.describe(include='all') forces pandas to include **all column types** in the summary. Missing review ratings were replaced with the median rating within each product category.

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[11]: df.describe(include='all')
```

[11]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

- Column names were standardized into snake_case format.
- Created age_group variable to classify customers by age range.
- Converted purchase frequency into numeric day format.
- Removed redundant promo_code_used column after validation.
- Cleaned data was uploaded into MySQL for SQL-based analysis.

4. Business Analysis (MySQL)

1. Revenue Analysis:

- Compared total revenue by gender.

gender	revenue
Male	157890
Female	75191

- Analyzed revenue contribution by age group.

age_group	groupwise_revenue
Young Adult	62143
middle-aged	59197
Adult	55978
Senior	55763

- Evaluated subscriber vs non-subscriber spending.

subscription_status	total_customer	average_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

2. Discount Impact:

- Determined products with highest discount usage.

item_purchased	purchase_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

- Identified high-spending discount users.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91

3. Product Performance:

- Ranked top 5 products by average rating.

item_purchased	avg_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

- Listed top 3 products per category.

rank_no	category	item_purchased	customers
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

4. Shipping & Segmentation:

- Compared Standard vs Express shipping purchase amounts.

avg (purchase_amount)	shipping_type
60.4752	Express
58.4602	Standard

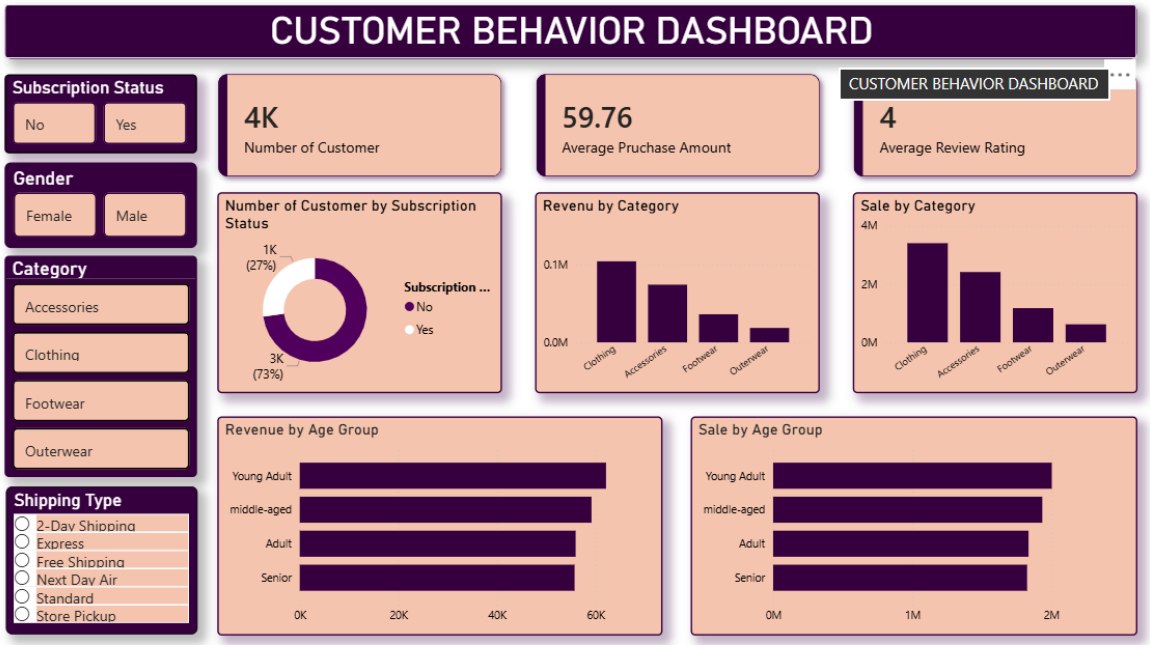
- Segmented customers into New, Returning, and Loyal groups.

segment_previous_purchase	segment
83	new
701	returning
3116	loyal

- Analyzed subscription likelihood among repeat buyers.

customers	subscription_status
958	Yes
2518	No

5. Power BI Dashboard



An interactive Power BI dashboard was developed to visualize revenue trends, customer segments, product performance, shipping behavior, and subscription impact. This tool allows stakeholders to explore insights dynamically.

6. Strategic Recommendations

- Expand subscription programs with exclusive benefits.
- Develop loyalty programs to reward repeat customers.
- Optimize discount policies to maintain profitability.
- Promote high-performing and top-rated products.
- Use targeted marketing for high-revenue segments.